

Tunisia, audited — the complete audit

Five absences. Nine legs. One metabolism. Every number recomputed in code, every correction applied into the design, every claim graded.

Provenance law: MEASURED = cited anchor · DERIVED = arithmetic on measured values · ASSUMED = declared default. commercially-validated = 0 — no buyer has signed anything; published as zero on purpose. Generated 2026-09-09 by nations_audit.py / make_nations_pdfs.py — the same code that renders the page.

1. The five absences and the one metabolism

Five structural absences — monetary sovereignty (debt \$40.5B, ~79% of GDP, 26.2% of export revenues to service), sovereign compute (InstaDeep: founded in Tunis, sold for \$680M), strategic materials (phosphogypsum at Gabès costing the sea >\$58M/yr), water, nutrition — answered by nine cash-positive legs on one metabolism of power, water, data and workforce. The audit is finished; the corrections are in the design.

2. The FX bridge — every dollar, by leg

Each leg below is volume × price × utilization, with the double-counting checks applied — the design carries the checked ledger.

LEG	TYPE	VOLUME	PRICE	\$M/YR	GRADE
Strategic metals (U + heavy REE, acid stream)	gross export	716,500 lb U3O8 + 585 t REO	\$80-95/lb U3O8 · \$30/kg REO	\$75-86M	STRONG
Solar fuel substitution (500 MW)	import substitution	745 GWh/yr displaced (876 GWh less 131 GWh to H2)	\$142/MWh (gas \$12/MMBtu x 11 MMBtu/MWh + \$10/MWh O&M); government anchor: 1,100 GWh ≈ \$125M/yr gas saving for the same 500 MW fleet (Pg14)	\$95-116M	PLAUSIBLE
Hydrogen → ammonia (loop-internal offtake)	import substitution	2.63 kt H2/yr → 14.9 kt NH3 for the GCT chain	\$500/t imported ammonia displaced (external offtake upside: \$8-9.5/kg H2)	\$7-25M	SPECULATIVE
Terfas (desert truffles)	gross export	940 t/yr mid (2,500 ha x 376 kg/ha; 70% exported)	EUR20-40/kg farm-gate	\$14-29M	PLAUSIBLE
Food import substitution	import substitution	1% of the \$3.0B food import bill	imported protein \$15-25/kg vs local \$5-10/kg	\$30-30M	SPECULATIVE

LEG	TYPE	VOLUME	PRICE	\$M/YR	GRADE
		(frame; volume bridge is a pilot deliverable)			
Compute colocation services	gross export (services)	colo park, tenant-dependent	market colocation rates	\$15-15M	<i>SPECULATIVE</i>
ESG financing delta	financing delta (not FX)	100-150bp on ~\$800M portfolio debt	interest saved, not foreign exchange	\$8-12M	PLAUSIBLE
Total external-account effect (mid)				~\$279M/yr	\$80M strong · \$137M plausible · \$61M speculative · \$0 verified

By kind: exports \$117M · import substitution \$152M · financing delta \$10M (interest saved, honestly labeled as not-FX). **By grade:** \$80M strong · \$137M plausible · \$61M speculative · \$0 verified — nothing is commercially validated yet, and that ledger reads zero on purpose.

3. The energy ledger — every kWh, by leg

LEG	GWH/YR	THE FORMULA
Pump-as-turbines (return flows)	3.9	$70\% \times 150\text{k m}^3/\text{d} \times 50\text{ m} \times 75\% \times 8,760\text{ h}$
Heat (kiln + exotherm)	30	9% of 333 GWh envelope
Flexibility (load shift)	20	dispatch value — not recovered joules
CO ₂ ; mineralized	30 kt	mass, not energy
Total	33.9 recovered + 20 dispatch	isobaric ERD inside the 4 kWh/m³ benchmark — never double-counted

4. Phase 0 — every dollar, itemized

ITEM	COST	WHAT IT BUYS
Water panel: 100 points, full suite (F, Cd, Pb, Cr, As, salinity, nitrate)	\$35,000	the first anchor measurement
Acid-stream assays: 12 points (U, REE, Ra-226, Po-210, full metals) + GCT access	\$40,000	the second anchor measurement
Emissions inventory: 30 passive samplers (PM _{2.5} /SO ₂ /fluorides) + lab	\$25,000	baseline for the health scoreboard
Cohort protocol: 3,000-family registry, ethics, data platform v0	\$35,000	the DALY ledger becomes MEASURED

ITEM	COST	WHAT IT BUYS
Health pilot first tranche: 100 households + 2 schools, 90 days	\$58,000	filters, clean-air boxes, KPIs
Legal + finance: health fund, industrial SPV, land-lease pre-agreements, tender registration	\$25,000	structure before scale
Engineering + project management	\$35,000	the 90 days are run
List price	\$253,000	list price \$253k; the two anchor measurements cost \$75k and land first — the claimed range is the phased-execution plan, not the list

5. Every claim, graded

CLAIM	GRADE	WHY
\$40.5B debt (~79% GDP); 26.2% of export revenues to service	STRONG	MEASURED via World Bank API on 2026-09-09: debt \$40.46B, debt/GDP 78.7%, service 26.17% of export revenues (2024)
\$3.0B food and \$4.9B fuel imports	STRONG	fuel \$4.895B verified on WITS/Comtrade (HS27, 2024); food per the WB food classification
InstaDeep founded in Tunis, sold for \$680M	STRONG	public deal record; the emblem for talent without infrastructure
Gabès: >\$58M/yr of REE value lost to the sea	STRONG	Science of the Total Environment (2021): up to 1,523.67 t/yr REE, ~\$58M/yr losses
U + heavy REE from the acid stream, ~\$80M/yr	STRONG	flowsheet industrial since 1978 (Freeport), patent expired, Morocco building it now; volume derived from cited primitives; Gafsa-specific assays are Phase 0
~\$285M/yr external-account effect, tiered	TIERED	\$80M strong (proven flowsheet/cited anchor) · \$137M plausible (gated) · \$68M speculative (assumed volume/price)
33.9 GWh/yr recovered + 20 GWh/yr dispatch value	DERIVED	every leg computed from cited primitives; dispatch value honestly labeled — not joules
\$2,592 per healthy year (water rung)	PLAUSIBLE	arithmetic DERIVED in code; DALY anchors transferred from Iranian studies — the cohort baseline makes it MEASURED
Phase 0 = \$150-250k, itemized	PLAUSIBLE	itemized list price \$253k; the two anchor measurements cost \$75k and land first
Commercially validated	ZERO	no buyer has signed anything — published as zero on purpose

6. The hidden dependencies

DEPENDENCY	TYPE	STATUS	NOTE
Desal brine permit	regulatory	in design	Gabes outfall / diffuser; permit required before SWRO FID

DEPENDENCY	TYPE	STATUS	NOTE
PG chemistry (Ra-226 partitioning)	technical	Phase 0 assay	flowsheet keeps radium in gypsum (Freeport precedent); re-verify on Gafsa stream
Uranium handling	regulatory	gate	CNSTN licensing, NORM transport, export controls
REE separation capex	capital	later phase	+\$244M solvent-extraction plant beyond the ~\$100M circuit
Solar bankability	financial	gate	DSCR ~0.9-1.0 at the record-low tariff; bankable region or grant support required
Grid connection	infrastructure	later phase	3 GW connection + 400 kV upgrades ~\$450M
Compute accelerators	regulatory	gate	export-control licenses; colocation until an anchor tenant is licensed
Truffle yield	agronomic	gate	376 kg/ha is the 20-yr Murcia average; gate = 3-season record ≥ 200 kg/ha
Export prices	market	exposure	U spot \$80-95/lb; truffle EUR20-40/kg — no contracted offtake yet
Financing	financial	uncommitted	DFI channel modeled for solar/desal; nothing committed

7. The structure and the pilot/capital separation

The design is six independently bankable modules sharing infrastructure. The loop is the outcome, never the precondition — no module's economics depends on another module succeeding, and every gate is public. World Bank CCDD (2023): Tunisia's water-sector investment need ~\$17.1B and energy-sector ~\$34.7B through 2050. The page is not that program and does not claim to be: Phase 0 is \$150-250k of measurements, and the whole core build is ~\$250-650M phased over years 2-5, module by module, each gated. The CCDD numbers confirm the scale of the problem; they do not compete with a pilot gate.

Africa Energy Portal (2026): 500 MW of utility-scale solar approved (~\$400M investment). Same class as the page's P3 module (500 MW) — the record-low ~\$31/MWh awards are the reference the DSCR gate is tested against.

8. The Fed transmission — measured, not restated

World Bank API, measured 2026-09-09 (Pg1): external debt **\$40.46B, 78.7% of GDP**; debt service **26.2% of export revenues**; reserves **\$9.34B** = 4.3 months of imports; food \$3.00B + fuel \$4.90B of imports; CA -1.5% of GDP; growth 2.5%, inflation 5.2% (2025).

Channel 1 — the debt trap: the “original sin” (Pg12) — no long-term borrowing in the dinar abroad — exposes the state to the Fed cycle (Pg13): Fed hikes lift global yields, refinancing turns expensive, and the 26.2% of export revenues that debt service absorbs compounds while reserves (4.3 months) drain.

Channel 2 — imported inflation: the rate differential pulls capital to US safe assets, the dinar depreciates, and the dollar invoice on food (\$3.00B) and fuel (\$4.90B) inflates. The BCT — autonomous, non-monetizing under Organic Law 2016-35 (Pg4/Pg5) — must hold rates to defend the dinar: correct orthodoxy, strangled growth.

The loop's answer: owned export engines earn ~\$117M/yr of gross exports outside the Fed cycle; the substitution legs take \$152M/yr out of the dollar invoice directly; reserves rise, the rating runs B-/Caa1 → BB-, and the refinancing premium shrinks for the state with its own hard-currency engines. Not defiance of the Fed — independence from the channel.

What this document is: a concept-level engineering design whose numbers are computed in code from sourced primitives — recomputed, corrected, and applied into the design. Free and public.

What it is not: a commissioned state program, a bank-grade feasibility study, an investment offer, or a claim that any of this has been built. Nothing here is advice to any government or any investor. Nothing replaces geological, radiological, geotechnical or EPC-level surveys. Verify every figure and citation before relying on any of it.

Works cited

- [Rm1] Rare Earth Elements in Phosphate Ores and Industrial By-Products (Gafsa avg 322 mg/kg, peaks >1,700) — <https://www.mdpi.com/2075-163X/15/12/1232...>
- [Rm2] Global phosphogypsum: generation, stockpiles and 60-85% REE partition to PG — <https://www.mdpi.com/2071-1050/17/19/8828...>
- [Rm3] Gabès phosphate plants REE characterization: enrichments to 1,075 mg/kg; Ce 80-309 mg/kg; negligible Dy/Tb — https://www.researchgate.net/publication/352414505_Rare_earth_elements_characterization_associated_to_the_phos...
- [Rm4] Characterization and Leaching Kinetics of REE from Phosphogypsum in HCl (1.65 mol/L; 65.6% ΣREE; 73.8% Y) — https://www.researchgate.net/publication/361001660_Characterization_and_Leaching_Kinetics_of_Rare_Earth_Elemen...
- [Rm5] Recovering REE from phosphogypsum as a secondary resource: scale-up failure modes (filtration, silica gel, co-extraction) — <https://www.chinanhd.com/recovering-rare-earth-elements-from-phosphogypsum-as-a-secondary-resource-pathway/...>
- [Rm6] PhosAgro pilot-scale REE recovery from apatite-derived PG (5 m³/h — igneous, richer feed) — [https://cris.vtt.fi/en/publications/pilot-scale-recovery-of-rare-earths-and-scandium-from-phosphogyps/...](https://cris.vtt.fi/en/publications/pilot-scale-recovery-of-rare-earths-and-scandium-from-phosphogyps/)
- [Rm7] Rainbow Rare Earths Phalaborwa: 2.2 Mtpa PG, 29.1% NdPr, pilot proven, OPEX <\$13/kg (exception, not the rule) — <https://www.rainbowrareearths.com/wp-content/uploads/2022/03/21-08-04-Sprott-Initiation.pdf...>
- [Rm8] 2024-2026 oxide prices: NdPr \$64-103/kg; Dy ~\$340/kg; Tb ~\$1,800-1,875/kg; La \$3.70/kg; Y <\$5-10/kg; Ce ~zero — <https://en.institut-seltene-erden.de/aktuelle-preise-von-seltenen-erden/...>
- [Rm9] Mixed rare-earth concentrate discount vs separated oxides (30-50%) — <https://pubs.acs.org/doi/pdf/10.1021/acssusresmg.5c00353...>
- [Rm10] Aclara: solvent-extraction separation plant Class-5 CAPEX \$244M, OPEX \$12/kg REO — <https://investingnews.com/aclara-announces-update-on-its-rare-earths-separation-project/...>
- [Rm11] Tunisian PG radiological characterization: Ra-226 214-970 Bq/kg; U-238 88.5; Pb-210 755.4; co-precipitation risk — https://www.researchgate.net/publication/24188010_Characterization_of_phosphogypsum_wastes_associated_with_phos...
- [Rm12] IAEA / EU BSS: 1 Bq/g (1,000 Bq/kg) exemption threshold for U-238/Ra-226; TENORM classification — <https://www.mdpi.com/2071-1050/17/8/3473...>
- [Rm13] NORM disposal options and costs (\$15-420 per 55-gallon drum) — <https://www.ogj.com/home/article/17230971/norm-disposal-options-costs-vary...>
- [Rm14] China's 85-90% share of global REE separation/refining capacity — <https://www.adlittle.com/sites/default/files/reports/ADL%20The%20rare%20earth%20imperative%202025.pdf...>
- [Rm15] Breakeven economics for non-Chinese REE projects (~\$77/kg NdPr-equivalent; PG-grade requires >\$150/kg) — <https://www.elibrary.imf.org/view/journals/001/2026/072/article-A001-en.xml...>
- [Rm16] Hydrochloric acid prices: EU \$124-150/t (Q4 2025); NA ~\$350/t; China ~\$15/t — <https://chemtradeasia.sg/en/market-insights/hydrochloric-acid-supply-report-2024-2025-buyer-lessons...>
- [Rm17] GCT throughput (~6.5 Mt phosphate rock/yr; >10 Mt/yr PG at peak; 'hundreds of millions of tons' discharged at Gabès) — http://www.gct.com.tn/fileadmin/user_upload/Downloads/Rapport_Annuels/GCT-ANNUAL-REPORT-2012-PART1.pdf...
- [Rw1] Largest SWRO plants: Ras Al-Khair 1,036,000 m³/d; Taweelah 909,200 (\$847M-1.2B); Sorek II 672,000; Rabigh 3 600,000 — <http://www.ide-tech.com/en/project/sorek-b-desalination-plant/...>
- [Rw2] SWRO CAPEX \$800-1,200 per m³/day for mega-plants (>100k m³/d) — <https://supra-international.com/insights/seawater-reverse-osmosis-swro-desalination-technology-comprehensive-a...>
- [Rw3] LCOW at plant gate: \$0.40-0.60/m³ (Jubail 3A \$0.41; Sorek II bid \$0.32-0.45); energy share 40-60% of OPEX — <https://pmc.ncbi.nlm.nih.gov/articles/PMC8953854/...>
- [Rw4] Pumping physics: 400 m lift = 1.09 kWh/m³ at 100% eff; 1.45 at 75%; friction 0.5-1.5 kWh/m³ over 300 km; realistic 2.0-2.5 kWh/m³ — https://www.engineeringtoolbox.com/water-pumping-costs-d_1527.html...

22. [Rw5] Jubail-Buraydah IWTP: 587 km, 650,000 m³/d, \$2.26B (\$3.85M/km); twin-pipe scale \$5-7M/km — <https://www.meed.com/saudi-arabia-awards-22bn-water-transmission-contract...>
23. [Rw6] Delivered inland desalinated cost \$1.10-1.50/m³ (plant gate + transport + pumping energy) — <https://investriyadh.ai/intelligence/water-scarcity-investment/...>
24. [Rw7] Tunisian water tariffs: SONEDE 0.200-2.310 TND/m³; irrigation 0.060-0.160 TND/m³; groundwater 0.112-0.162 TND/m³ — <https://www.sonede.com.tn/tout-savoir-sur-leau/la-facture-deau-methodes-de-paiement/...>
25. [Rw8] International transfer precedents: GMMR \$33B (2.5-6.5 Mm³/d, \$0.28-0.62/m³, fossil water); Riyadh leveled transport \$0.42/m³; <\$0.20/m³ is the agronomic ceiling — <https://www.mei.edu/publication/whats-next-libyas-great-man-made-river-project/...>
26. [Rw9] 24/7 firming: 3.03 GW @20% CF = 0.606 GW avg (zero margin); 9.15 GWh/night; BESS round-trip 85% → 12-15 GWh; \$110-125/kWh → ~\$1.5B; solar+storage LCOE \$100-130/MWh — <https://www.energy-storage.news/battery-storage-system-prices-continue-to-fall-sharply-bnef-and-ember-reports-...>
27. [Rw10] Tunisia grid: STEG 5.9 GW, ~19 TWh/yr, 95% gas-fired, renewables <5%, industrial tariff 0.352 TND/kWh; budget deficit 5-7% GDP, debt ~83% GDP — <https://www.trade.gov/country-commercial-guides/tunisia-electrical-power-systems-and-renewable-energy...>
28. [Rw11] Agrivoltaics: 20-40% CAPEX premium over utility PV (+\$0.30/Wp); NREL range \$1.20-2.60/Wp — <https://www.mdpi.com/1996-1073/19/1/102...>
29. [Rw12] Tunisia land law: foreign ownership of agricultural land prohibited; 40-year leases only; state-domain conversion is an administrative hurdle — <https://2009-2017.state.gov/e/eb/efd/2008/101021.htm...>
30. [Rw13] Grid constraints: 3 GW overwhelms local 400 kV; new lines \$1-2.5M/km; renewables <5% of generation; curtailment risk — <https://reglobal.org/tunisia-focuses-on-grid-expansion-for-integrating-renewable-energy/...>
31. [Rc1] Underground DC precedents: Lefdal Mine (olivine; 200 MW; PUE 1.08-1.15 via 8°C fjord water); SubTropolis limestone (18°C + mechanical cooling); no phosphate-mine DC exists — <https://www.lefdalmine.com/...>
32. [Rc2] Gafsa basin: DInSAR-documented subsidence at Moulares; geothermal gradient 25-30°C/km → 23-29°C at 100-300 m depth — https://www.researchgate.net/publication/251701016_Phosphate_mine_subsidences_deduced_from_differential_interf...
33. [Rc3] Gafsa radiation: 226Ra 16.3-375.1 Bq/kg in soils; phosphate tailings to 7,606 Bq/kg; radon-222 accumulation in enclosed workings; alphas-induced soft errors — https://www.researchgate.net/publication/380540277_Assessment_of_radiation_exposure_in_the_phosphate_mining_ar...
34. [Rc4] ASHRAE TC9.9 Class H1 (5th ed.): 18-22°C recommended, 25°C max for AI accelerators; AI datacenter CAPEX \$20-35M/MW (incl. IT), \$8-15M/MW facility-only — <https://envigilance.com/compliance/ashrae-tc-9-9/...>
35. [Rc5] GPU cost: ~100,000 B200-class per 100 MW; \$25-40k each → \$2.5-4B per 100 MW; 300 MW ≈ \$7.5-12B — <https://intuitionlabs.ai/articles/data-center-gpu-pricing-2026...>
36. [Rc6] US BIS export controls 2024-26: UAE 35,000 GPUs (G42/HUMAIN, 'Compute Diplomacy'); Saudi case-by-case 35,000 cap; Tunisia has no bilateral AI-security agreement — <https://www.commerce.gov/news/press-releases/2025/11/statement-uae-and-saudi-chip-exports...>
37. [Rc7] Terfezia cultivation: commercial since 1999-2001 (Murcia, Spain); fruiting at year 2-3; 20-yr studies: average ~376 kg/ha, peak 1,069 kg/ha, drought years 0-2 kg/ha — <https://pmc.ncbi.nlm.nih.gov/articles/PMC8907101/...>
38. [Rc8] Desert-truffle water 200-250 mm/yr; protein 14-22.5% dry weight (the page's '20-27%' sits at the top of the reported range) — <https://www.mdpi.com/2073-4395/11/8/1462...>
39. [Rc9] Terfezia farm-gate €20-60/kg (€200/kg only in extreme scarcity in Gulf markets); Tuber melanosporum €700-1,500/kg retail — different commodities — <https://trufamania.com/desert-truffles.htm...>
40. [Rc10] Mycoclature economics: black truffle payback 10-15 years; Terfezia fruits faster but at one tenth the price — <https://rovirosafarms.com/melanosporum/...>
41. [Rc11] Tunisia sovereign ratings: Moody's Caa1 (Feb 2025 upgrade, stable); Fitch B- (Sep 2025, affirmed Jan 2026); reserves ~4.5 months import cover; WACC 10-15% — <https://www.fitchratings.com/research/sovereigns/fitch-upgrades-tunisia-to-b-outlook-stable-12-09-2025...>
42. [Rc12] Who finances Tunisia: EIB, AfDB, KfW, World Bank/IFC, EU Global Gateway (17 GW renewable programme); no active IMF program — <https://international-partnerships.ec.europa.eu/policies/global-gateway/17-gw-renewable-energy-programme-tunis...>
43. [Rc13] Empirical IPP record: 500 MW solar awarded 2024/25 (Scatec, Qair, Voltaia 25-yr PPAs at ~\$0.031/kWh); installed renewables ~400 MW; STEG debt >\$1.3B — <https://www.engineeringnews.co.za/article/scatec-awarded-ppas-for-wind-solar-projects-in-tunisia-2026-01-26...>
44. [Rc14] Chain logic: 8 interdependent stages at a generous 70% individual success ⇒ ~5.8% joint probability (0.7⁸) — <https://www.mdpi.com/2073-4395/14/12/3021...>
45. [Rc15] Terfezia clavervyi salt sensitivity: establishes only in near-fresh soils, EC 0.82-0.85 dS/m — https://jabonline.in/abstract.php?article_id=1363&sts=2...
46. [Rc16] Terfezia plantation density: ~6,000 mycorrhized plants per hectare — https://www.researchgate.net/publication/299679495_Preparation_and_Maintenance_of_Both_Man-Planted_and_Wild_Pl...
47. [Rc17] Saudi Al-Khalidiah 150-ha Tirmania plantation: largely failed (14 kg/ha after soil import; rainfall collapse) — <https://ojs.openagrar.de/index.php/JABFQ/article/download/2184/2570...>
48. [Rc18] Desert-truffle market depth: Mamora (Morocco) wild harvest peaks ~100 t/yr; luxury band breaks past ~1,000-1,500 t/yr of new supply — <https://www.mdpi.com/1999-4907/14/5/952...>
49. [Rc19] Tunisia arid land: ~86,000 ha of state domain earmarked for 40-year agricultural leases (Law 93-120) — <https://documents1.worldbank.org/curated/en/926441468778187741/pdf/multiopage.pdf...>

50. [Ru1] Uranium from wet-process phosphoric acid: Freeport (US) produced 14M lbs U₃O₈ 1978-99 at >95% DEHPA/TOPO recovery; ~20% of US uranium in the 1980s came from phosphates — <https://www.osti.gov/servlets/purl/1357599...>
51. [Ru2] URANEXT/OCP (Morocco) building the first modern yellowcake-from-phosphoric-acid plant, ~\$100M CAPEX, El Jadida, 2025 — <https://northafricapost.com/87963-uranext-to-build-moroccos-first-yellowcake-uranium-plant-in-el-jadida.html...>
52. [Ru3] WPA uranium SX cost \$59-83/lb U₃O₈ (IX \$46-76) vs \$80-95/lb 2024-26 spot — <https://repositories.lib.utexas.edu/bitstreams/89897275-85cf-4cd6-a0c5-1d8daa353799/download...>
53. [Ru4] REE from phosphoric acid: PhosAgro pilot; D2EHPA extracts >85% of heavy REEs (Dy, Y, Er) from WPA — <https://www.mdpi.com/2673-4117/7/2/58...>
54. [Ru5] Gafsa phosphate rock uranium content 45-140 ppm (southern basin) — https://www.researchgate.net/publication/380540277_Assessment_of_radiation_exposure_in_the_phosphate_mining_ar...
55. [H1] Guissouma 2017 (Chemosphere): Tunisia tap-water fluoride 0-2.4 mg/L; 25% of population at dental-fluorosis risk, 20% at skeletal-fluorosis risk (WHO limits) — <https://pubmed.ncbi.nlm.nih.gov/28284958/...>
56. [H2] Bouchiba 2026 (Toxics 14(5):438): Mdhilla beneficiation effluent Cd 0.49, Pb 0.71, Cr 3.09 mg/L — exceeding discharge limits; risk driven by Cd, Co, Tl — <https://doi.org/10.3390/toxics14050438...>
57. [H3] Gupta 1996 RCT: fluorosis REVERSED in children (Ca + VitD₃ + ascorbic acid; n=25; water 4.5 ppm F; double-blind) — <https://pubmed.ncbi.nlm.nih.gov/8942013/...>
58. [H4] Ghaemi 2024 (Iran): dental-fluorosis DALY rate up to 981.45 (UI 353.23-1618.40) per 100,000 in a high-F district; Monte Carlo dose-response — <https://pubmed.ncbi.nlm.nih.gov/39003868/...>
59. [H5] Abtahi 2019 (Water Research): age-sex specific DALYs from drinking-water fluoride, Iran national study — <https://pubmed.ncbi.nlm.nih.gov/30953859/...>
60. [H6] Naddafi 2022 (Env Research): burden of disease from heavy metals in drinking water, Iran 2019 — <https://pubmed.ncbi.nlm.nih.gov/34529973/...>
61. [H7] Egner 2014 RCT (n=291): broccoli-sprout beverage 12 wks -> urinary benzene mercapturic acid +61%, acrolein +23% (P<=0.01); GSTT1-positive excrete more — <https://pubmed.ncbi.nlm.nih.gov/24913818/...>
62. [H8] Pabis 2018: zinc supplementation -> >30% reduction of kidney/liver Cd in aged metallothionein-transgenic mice — <https://pubmed.ncbi.nlm.nih.gov/30103140/...>
63. [H9] Zhai 2016: Lactobacillus plantarum CCFM8610 binds gut Cd, protects the intestinal barrier, lowers tissue Cd in mice — <https://pubmed.ncbi.nlm.nih.gov/27208136/...>
64. [H10] Mueller 2018: N95-class respirators >=98% filtration efficiency vs <=44% for any cloth — <https://pubmed.ncbi.nlm.nih.gov/29779694/...>
65. [H11] Allen & Barn 2020: HEPA purifiers substantially cut indoor PM_{2.5} and improve subclinical cardiopulmonary health; population benefits often exceed costs — <https://pubmed.ncbi.nlm.nih.gov/33241434/...>
66. [H12] Drieling 2022 RCT (75 asthmatic children): HEPA in bedroom+living room -> poorly-controlled asthma IRR 0.43-0.45; unplanned clinical utilization IRR 0.35 — <https://pubmed.ncbi.nlm.nih.gov/34980119/...>
67. [H13] Spirulina RCT (Cambodia, n=179): 2 g/day x 50 days -> anemia reduction significant (p=0.004); weight-gain trend p=0.07 (do not overclaim growth) — <https://pubmed.ncbi.nlm.nih.gov/36476193/...>
68. [H14] EJAtlas: Gafsa affected population 30,000-70,000; cancer/asthma/infertility reported high; conflict since 2008 — <https://ejatlas.org/conflict/phosphate-mining-in-gafsa...>
69. [H15] NRGi: 'serious dearth of rigorous studies' linking phosphate operations to health outcomes; 7% birth-defect claim is activist-reported, unverified — <https://resourcegovernance.org/articles/uncovering-impacts-phosphate-mining-tunisian-women...>
70. [H16] Geographical 2024: asthma onset age 2 documented (Redeyef); nearest hospital 70 km; doctors prescribe leaving Gafsa — <https://geographical.co.uk/science-environment/gafsas-silent-suffering-the-hidden-cost-of-phosphate-mining-in-...>